



Feedback form for the course

Please do fill this in before the end of term

We do take comments seriously, and they help to make the course better over time

https://cambridge.eu.qualtrics.com/jfe/form/SV_3z0oG7tq3O9xtpc

Mode splitting and coupling

From last lecture

Weak form of the equations of motion can be written:

$$\langle \mathbf{w} | P | \ddot{\mathbf{u}} \rangle + \langle \mathbf{w} | W | \dot{\mathbf{u}} \rangle + \langle \mathbf{w} | H | \mathbf{u} \rangle = \langle \mathbf{w} | \mathbf{f} \rangle,$$

Eigenvalue problem in non-rotating case:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

Eigenfunction expansion solution in frequency-domain:

$$\tilde{\mathbf{u}} = \sum_{km} \frac{\langle km | \tilde{\mathbf{f}} \rangle}{\omega_k^2 - \omega^2} |km\rangle,$$

Spherically symmetric Earth models

Angular momentum operators

Gradient in spherical polar co-ordinates:

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \frac{1}{r} \hat{\boldsymbol{\theta}} \frac{\partial}{\partial \theta} + \frac{1}{r \sin \theta} \hat{\boldsymbol{\phi}} \frac{\partial}{\partial \phi},$$

Used to define the **tangential gradient operator** on the unit sphere:

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \frac{1}{r} \nabla_1,$$

Associated operators related to those for angular momentum:

$$\hat{\mathbf{r}} \times \nabla_1 \propto \hat{\mathbf{L}}, \quad \nabla_1^2 \propto \hat{\mathbf{L}}^2$$

Spherical harmonics

Simultaneous eigenfunctions of commuting operators:

$$\nabla_1^2 Y_{lm} = -l(l+1)Y_{lm}, \quad \frac{1}{i} \frac{\partial Y_{lm}}{\partial \varphi} = mY_{lm},$$

The **degree**, l , is a non-negative integer

For degree, l , the **order** is an integer between $-l$ and l

Orthonormality condition:

$$\int_{S^2} Y_{lm}^* Y_{l'm'} dS = \delta_{ll'} \delta_{mm'}.$$

Complete basis for scalar functions on the unit sphere

Vector spherical harmonics

To work with vector fields, we defined:

$$\mathbf{A}_{lm} = Y_{lm} \hat{\mathbf{r}}, \quad \mathbf{B}_{lm} = \nabla_1 Y_{lm}, \quad \mathbf{C}_{lm} = -\hat{\mathbf{r}} \times \nabla_1 Y_{lm}.$$

Orthogonality relations:

$$\begin{aligned} \int_{\mathbb{S}^2} \mathbf{A}_{l'm'}^* \cdot \mathbf{A}_{lm} dS &= \delta_{ll'} \delta_{mm'}, \\ \int_{\mathbb{S}^2} \mathbf{B}_{l'm'}^* \cdot \mathbf{B}_{lm} dS &= l(l+1) \delta_{ll'} \delta_{mm'}, \\ \int_{\mathbb{S}^2} \mathbf{C}_{l'm'}^* \cdot \mathbf{C}_{lm} dS &= l(l+1) \delta_{ll'} \delta_{mm'}, \end{aligned}$$

Complete basis for vector fields:

$$\mathbf{u} = \sum_{lm} (U_{lm} \mathbf{A}_{lm} + V_{lm} \mathbf{B}_{lm} + W_{lm} \mathbf{C}_{lm}),$$

Eigenvalue problem

Eigenvalue in non-rotating case:

$$-\omega^2 \langle \mathbf{w} | P | \mathbf{s} \rangle + \langle \mathbf{w} | H | \mathbf{s} \rangle = 0.$$

If the model is spherically symmetric we can use **separation of variables**

Angular dependence expressed in terms of vector spherical harmonics

For each degree and order, we get a **radial ODE eigenvalue problem**

Huge amount of algebra and the need to work in detail with vector and tensor spherical harmonic expansions

We will just summarise the main results

Toroidal modes

Two classes of eigenfunctions emerge from the separation of variables

Toroidal modes have eigenfunctions of the form:

$$\mathbf{s}(r, \theta, \varphi) = W(r) \mathbf{C}_{lm}(\theta, \varphi),$$

Radial part, W , depends on degree but not order – results in $(2l+1)$ -dimensional eigenspaces, this being characteristic of rotationally invariant systems

Eigenfunctions vanish in fluid regions (analogous to S-waves)

Displacements tangential to spherical surfaces, gravity not involved in their dynamics

Spheroidal modes

Spheroidal modes have the form:

$$\mathbf{s}(r, \theta, \varphi) = U(r)\mathbf{A}_{lm}(\theta, \varphi) + V(r)\mathbf{B}_{lm}(\theta, \varphi),$$

Radial functions again independent of order – **(2l+1)-dimensional eigenspaces**

Eigenfunctions can be non-zero in both solid and fluid regions

Radial and tangential components to the displacement, self-gravity contributes to their dynamics

Toroidal eigenvalue problem

Spheroidal and toroidal modes are orthogonal due to properties of VSH

For toroidal modes of degree l and order m , we can take test functions:

$$\mathbf{w}(r, \theta, \varphi) = W'(r) \mathbf{C}_{lm}(\theta, \varphi)$$

It is then easy to show that:

$$\langle \mathbf{w} | P | \mathbf{s} \rangle = \zeta^2 \int_0^b \rho W'^* W r^2 dr, \quad \zeta = \sqrt{l(l+1)}$$

Potential energy form similarly reduced to radial integral, but calculation harder

Toroidal eigenvalue problem

Weak form of the eigenvalue problem:

$$\begin{aligned} & -\omega^2 \int_0^b \rho W'^* W r^2 \, dr + \int_0^b \mu \left(r \frac{dW'^*}{dr} - W'^* \right) \left(r \frac{dW}{dr} - W \right) \, dr \\ & + (\zeta^2 - 2) \int_0^b \mu W'^* W \, dr = 0, \end{aligned}$$

Equivalent to 2nd order ODE eigenvalue problem (or Sturm Liouville type)

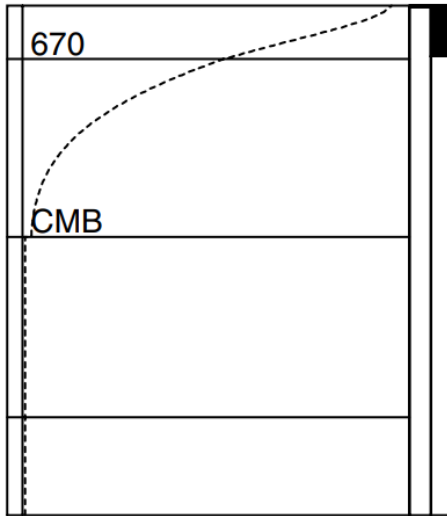
Closed-form solution in homogeneous planet using spherical Bessel functions

More generally, numerical methods needed

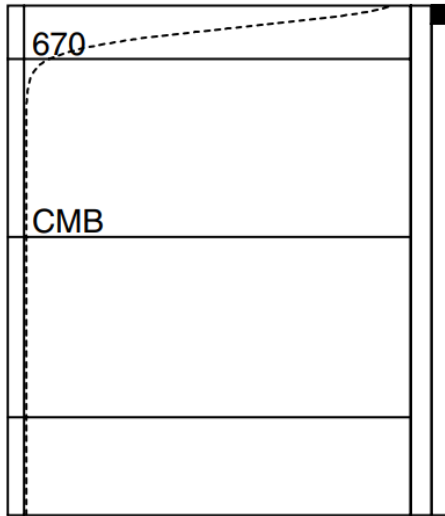
For each l , we get an infinite set of **overtones** labelled by an integer, $n = 0, 1, 2, \dots$

Toroidal eigenfunctions

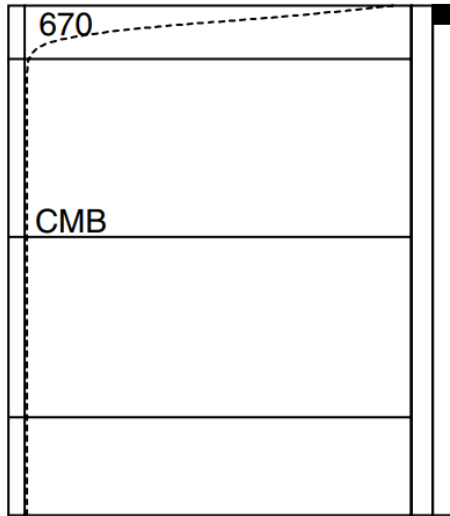
${}_0T_{10}$ Period = 619.9s $Q = 173.2$
Surface Eigenfunction



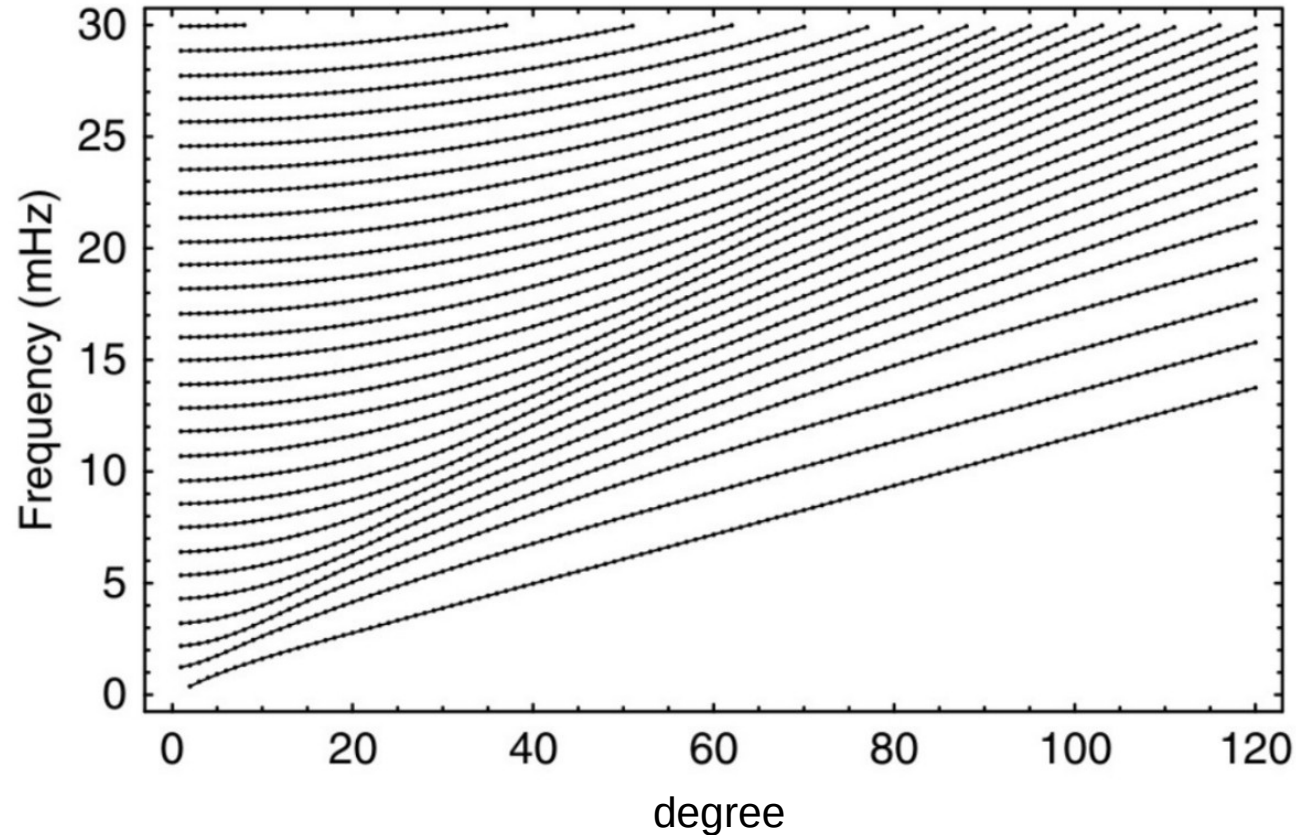
${}_0T_{50}$ Period = 164.6s $Q = 131.1$
Surface Eigenfunction



${}_0T_{100}$ Period = 86.5s $Q = 139.2$
Surface Eigenfunction



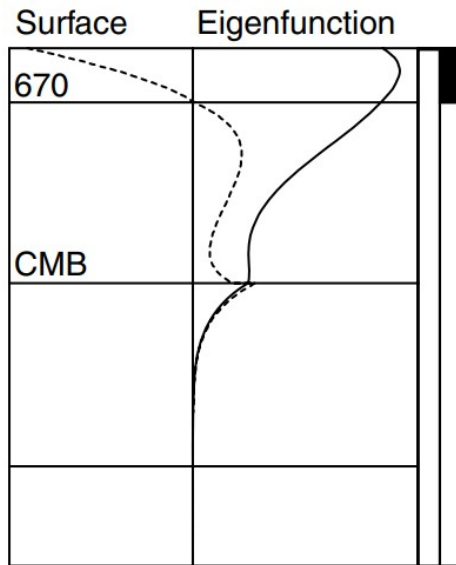
Toroidal mode eigenfrequencies



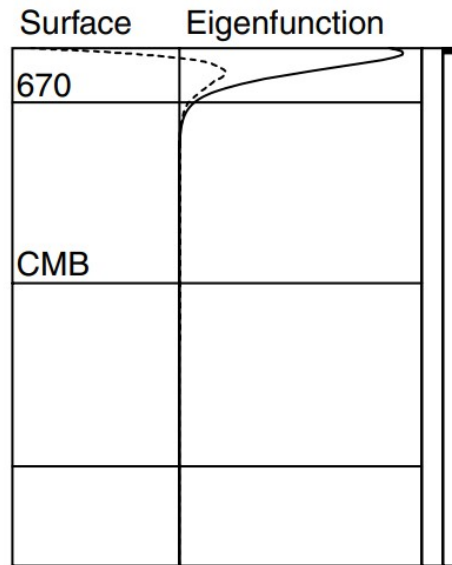
Woodhouse & Deuss (2007)

Spheroidal mode eigenfunctions

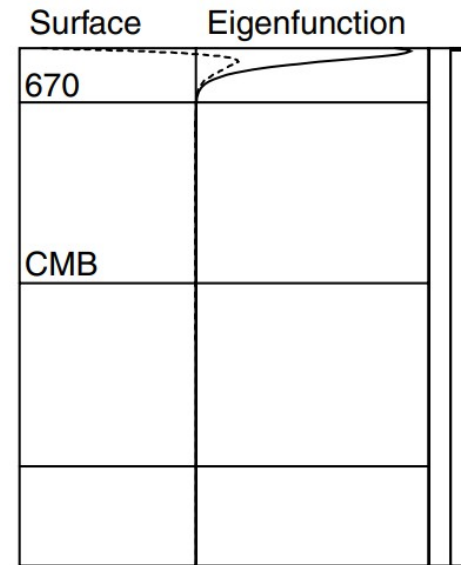
${}_0S_{10}$ Period = 579.2s Q = 327.8



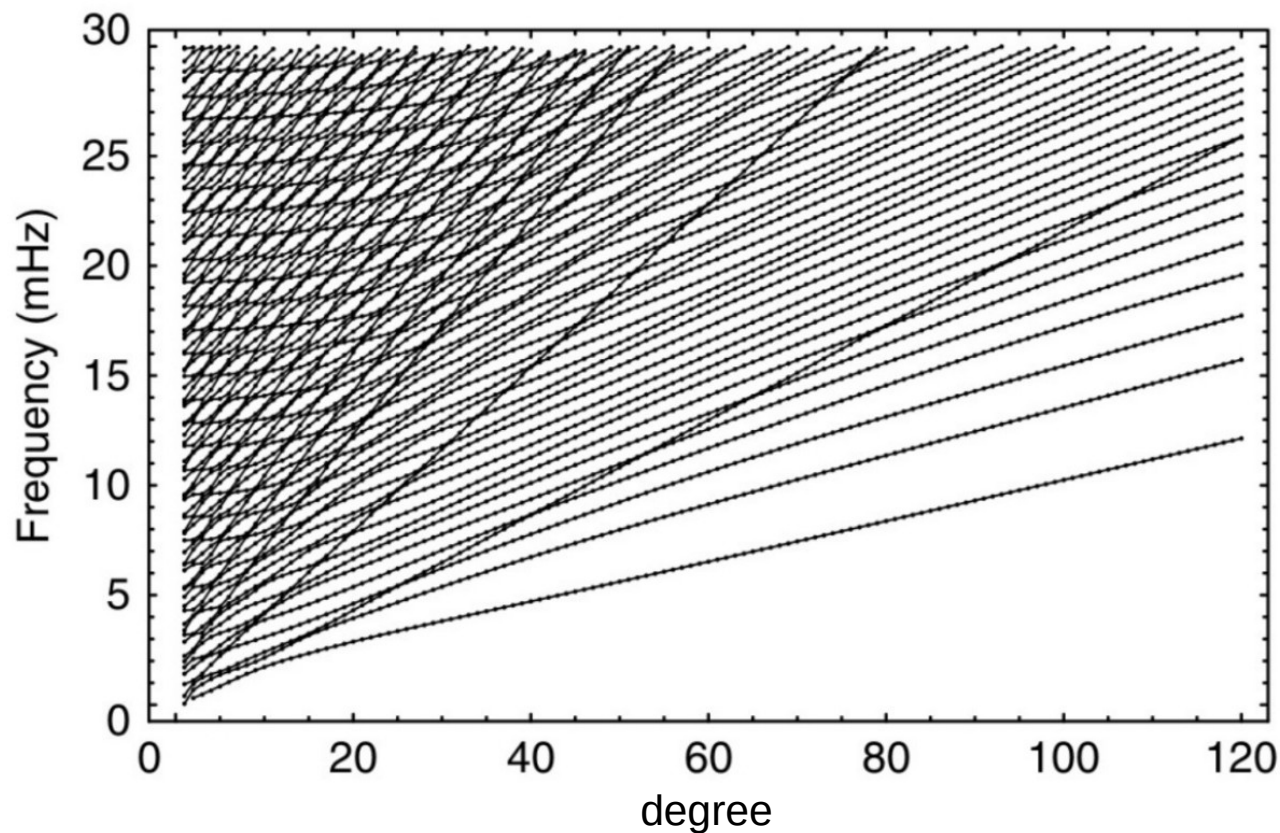
${}_0S_{50}$ Period = 178.2s Q = 143.2



${}_0S_{100}$ Period = 97.6s Q = 118.0



Spheroidal mode eigenfrequencies



Woodhouse & Deuss (2007)

Multiplets and singlets

Each eigenspace labelled by **three integers**:

- 1) **Mode type**, q , with 1 = spheroidal, 2 = toroidal
- 2) **Degree**, l , a non-negative integer
- 3) **Overtone number**, n , this being a non-negative integer

We call each such eigenspace a **multiplet**

Within each multiplet of degree l there are then $(2l+1)$ eigenfunctions labelled by their order, m , lying between $-l$ and l

Each eigenfunction is known as a **singlet**

To denote singlets, we use the notations:

$$|qlnm\rangle, \quad |km\rangle$$

First-order splitting theory

Perturbed eigenvalue problem

Suppose we have solved the spherical reference problem:

$$-\omega_k^2 \langle \mathbf{w} | P_0 | km \rangle + \langle \mathbf{w} | H_0 | km \rangle = 0,$$

We consider lateral heterogeneity as a first-order perturbation:

$$-\omega^2 [\langle \mathbf{w} | P_0 | \mathbf{s} \rangle + s \langle \mathbf{w} | P_1 | \mathbf{s} \rangle + \dots] + [\langle \mathbf{w} | H_0 | \mathbf{s} \rangle + s \langle \mathbf{w} | H_1 | \mathbf{s} \rangle + \dots] = 0.$$

Following degenerate perturbation theory, we look for solutions of the form:

$$\omega = \omega_k + s \delta \omega + \dots$$

$$\mathbf{s} = \sum_{m=-l}^l a_m |km\rangle + s \delta \mathbf{s} + \dots$$

Splitting of the degeneracy

Following the standard argument, the calculation is reduced to solution of an $(2l+1)$ -dimensional Hermitian eigenvalue problem

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

This gives $(2l+1)$ values for the perturbed eigenfrequencies along with a zeroth-order eigenfunction for each

Generically, the degeneracy will be removed, with the multiplet **splitting** into $2l+1$ distinct eigenfrequencies

This is just like the **Zeeman effect** for the hydrogen atom

Rotational splitting

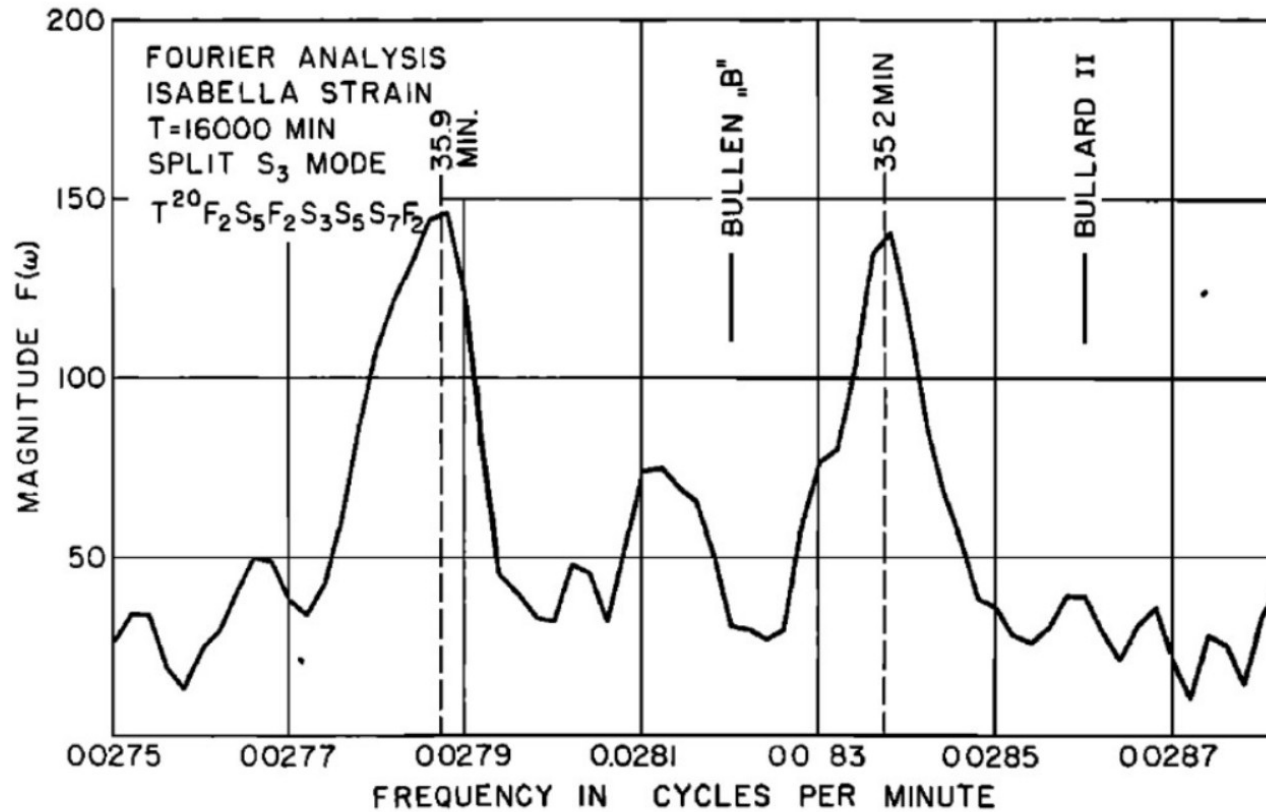
Rotational effects can be incorporated to first-order accuracy in the same way:

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

Various **selection rules** can be derived to determine the manner in which heterogeneity and rotation cause splitting

Extensions to **quasi-degenerate perturbation theory** are possible to account for coupling between two or more nearly degenerate multiplets

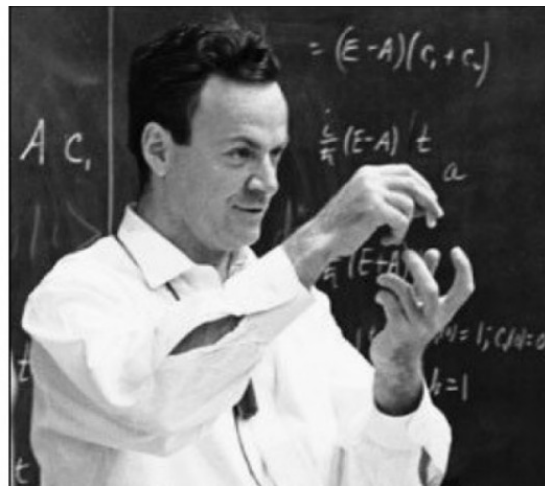
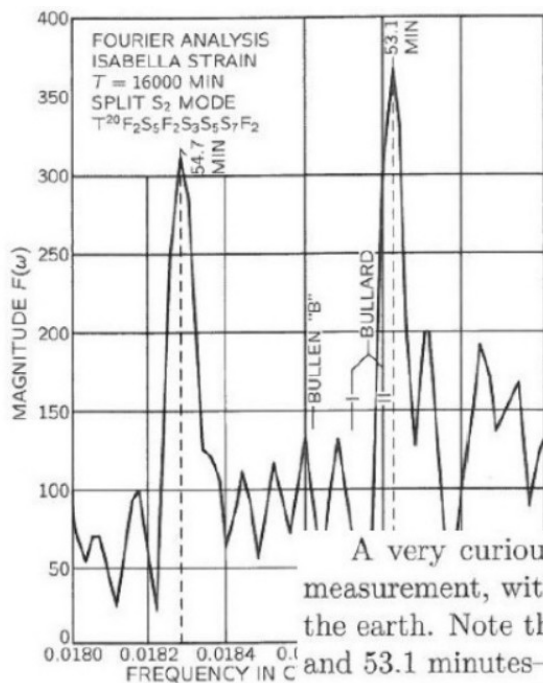
Spherically symmetric density



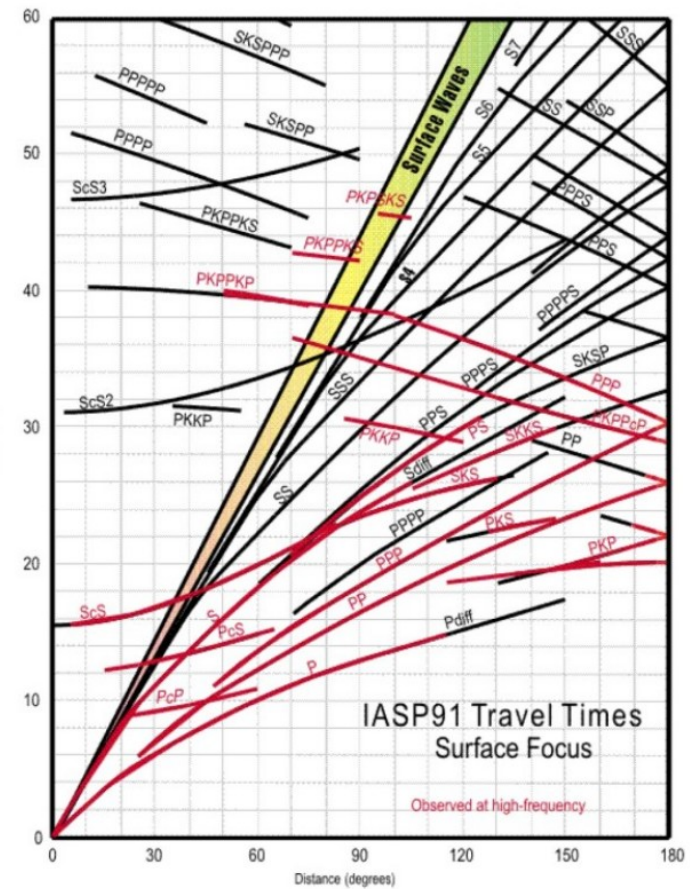
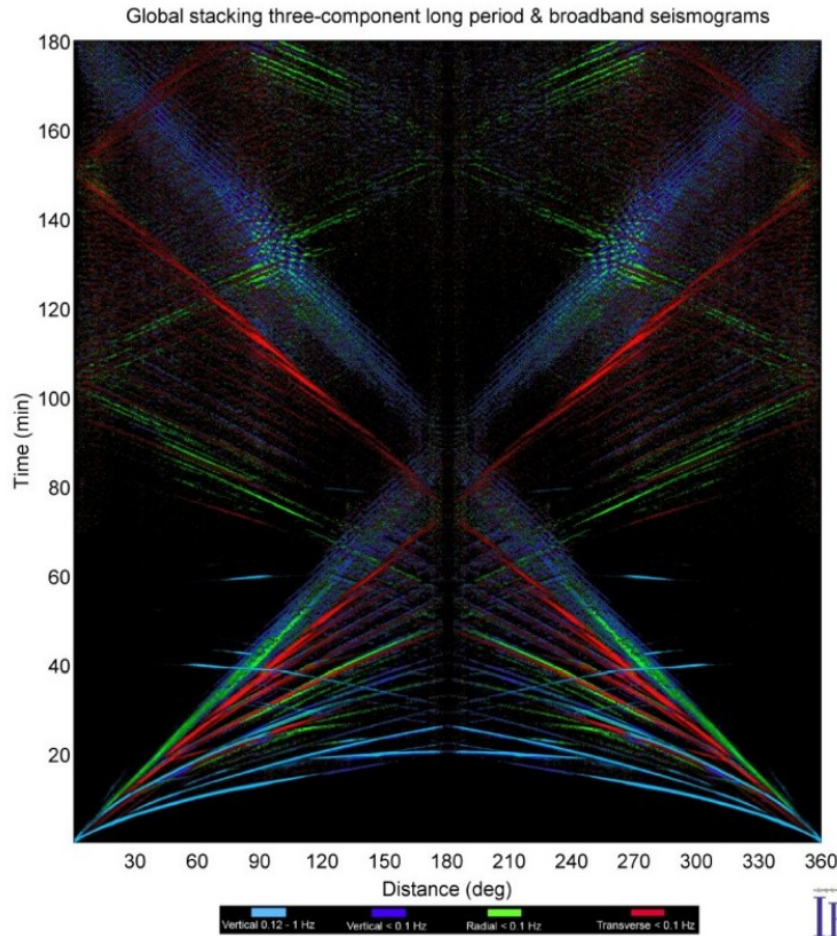
HUGO BENIOFF, FRANK PRESS, AND STEWART SMITH

FEBRUARY 1961

First observations of free oscillation spectra following the largest ever recorded earthquake



A very curious point is revealed in Fig. 51-8, which shows a very careful measurement, with better resolution of the lowest mode, the ellipsoidal mode of the earth. Note that it is not a single maximum, but a double one, 54.7 minutes and 53.1 minutes—slightly different. The reason for the two different frequencies was not known at the time that it was measured, although it may have been found in the meantime. There are at least two possible explanations: One would be that there may be asymmetry in the earth's distribution, which would result in two similar modes. Another possibility, which is even more interesting, is this: Imagine the waves going around the earth in two directions from the source. The speeds will not be equal because of effects of the rotation of the earth in the equations of motion, which have not been taken into account in making the analysis. Motion in a rotating system is modified by Coriolis forces, and these may cause the observed splitting.



Travel times for the same epicentral angle can be averaged to try to remove the effect of laterally varying structure prior to inversions

Diagonal sum rule

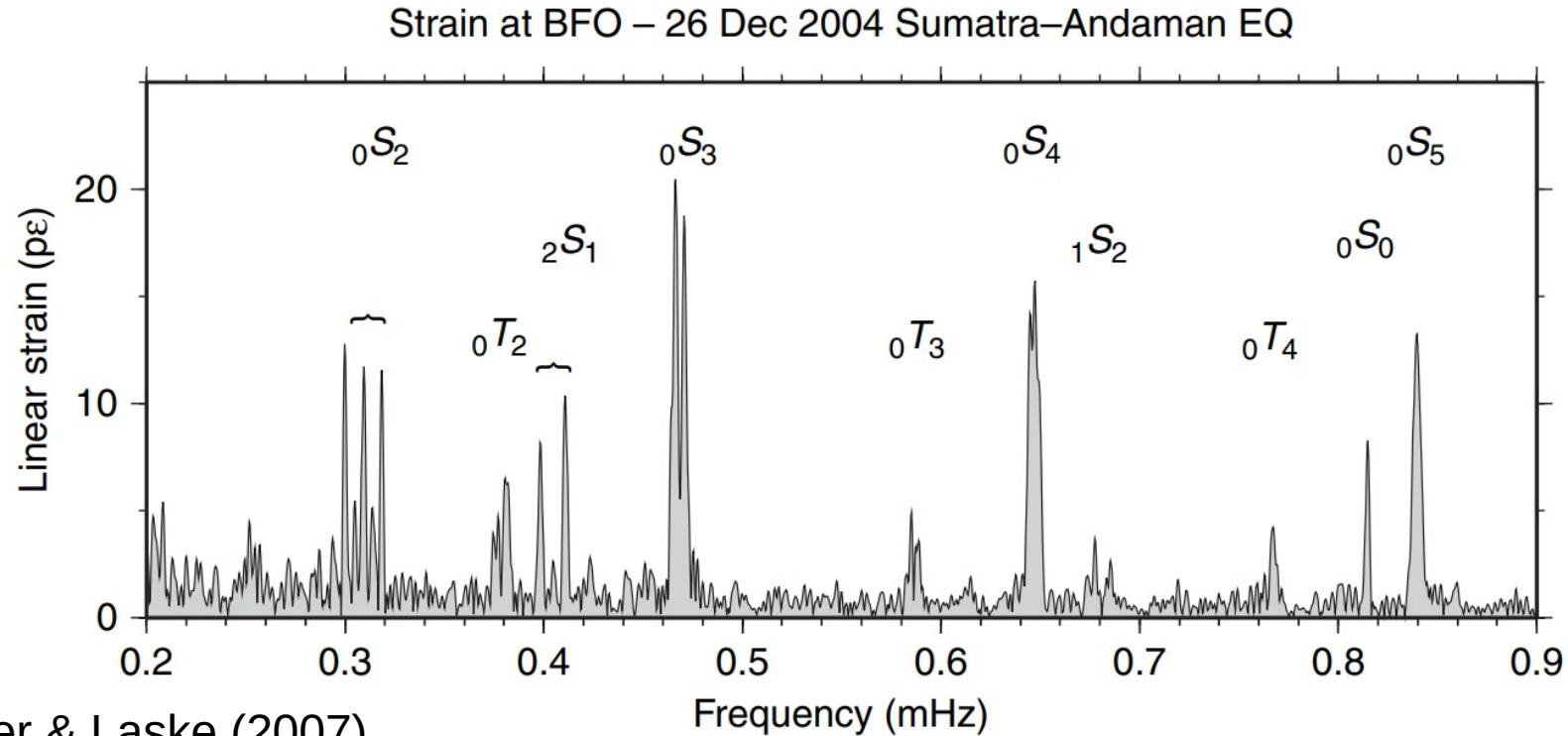
Eigenvalue problem associated with degenerate perturbation theory:

$$\frac{1}{2\omega_k} \sum_{m=-l}^l [-\omega_k^2 \langle km' | P_1 | km \rangle + i\omega_k \langle km' | W | km \rangle + \langle km' | H_1 | km \rangle] a_m = \delta\omega a_{m'}$$

Using symmetry arguments, it can be shown that the splitting matrix has zero trace – this implies that the eigenfrequency perturbations sum to zero

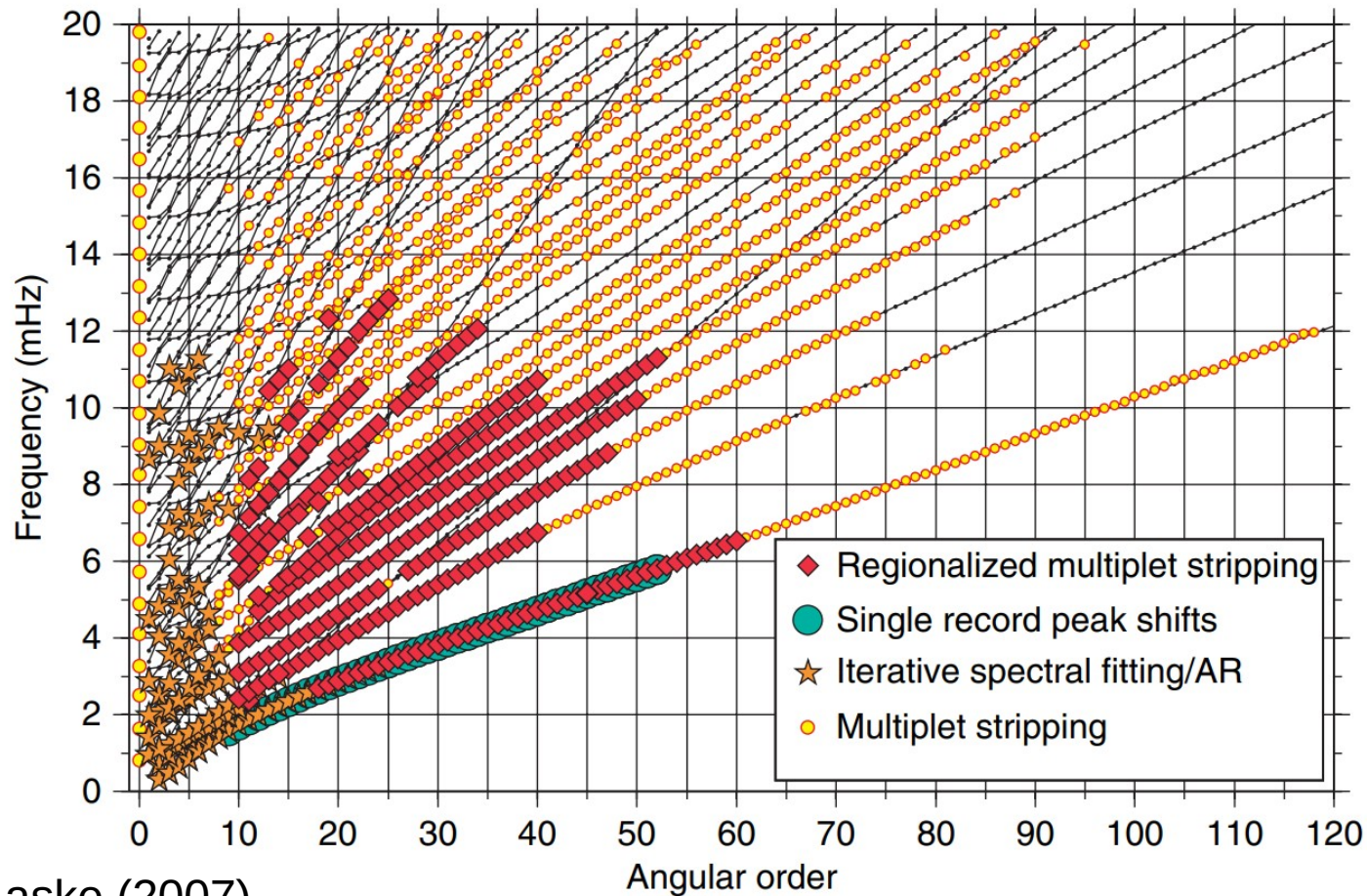
This is the **diagonal sum rule** due to Freeman Gilbert

It provides a rigorous basis for averaging observations of split eigenfrequencies within inversion for the spherically averaged structure



Widmer & Laske (2007)

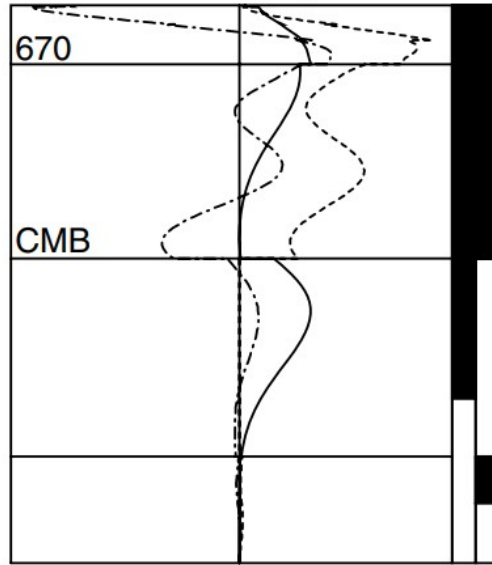
It is relatively rare that all split singlets for a multiplet can be identified and measured, but the diagonal sum rule still forms the basis for more sophisticated methods for measuring eigenfrequencies



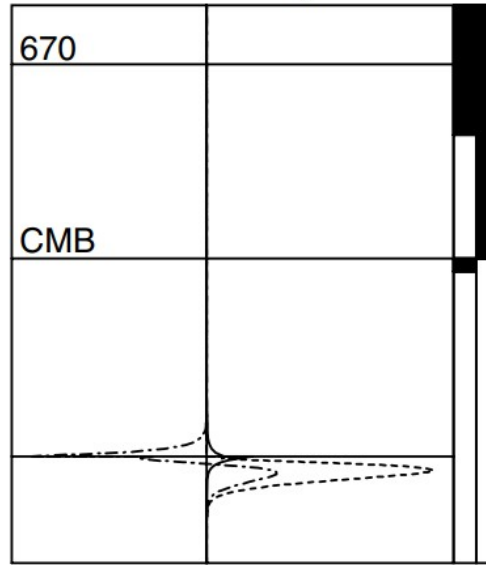
Widmer & Laske (2007)

By the late 70s, about 1000 eigenfrequencies had been measured

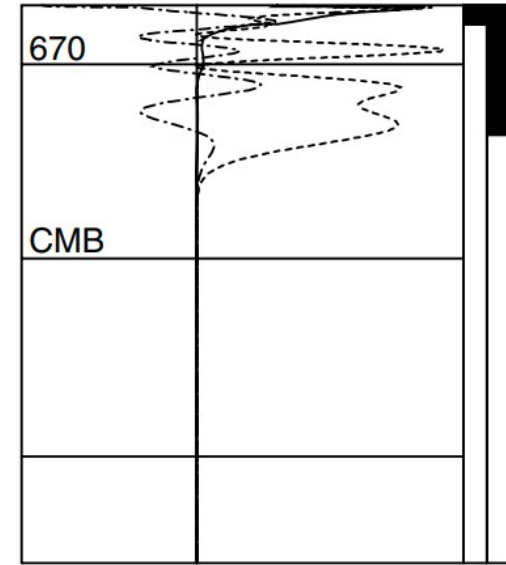
${}_5S_2$ Period = 478.2s Q = 317.9
Surface Sensitivity kernels



${}_5S_{10}$ Period = 240.6s Q = 89.5
Surface Sensitivity kernels

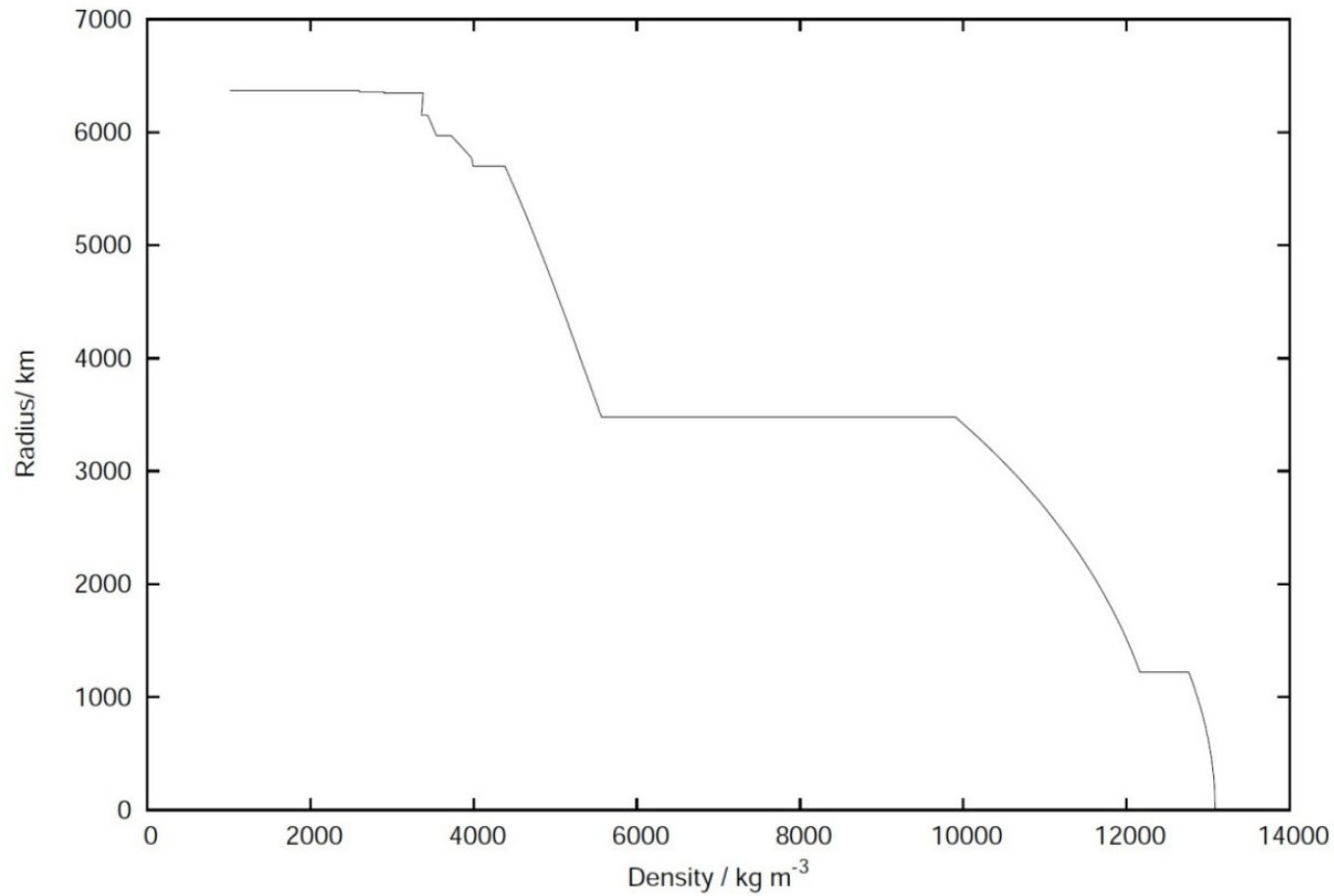


${}_5S_{50}$ Period = 91.1s Q = 241.9
Surface Sensitivity kernels



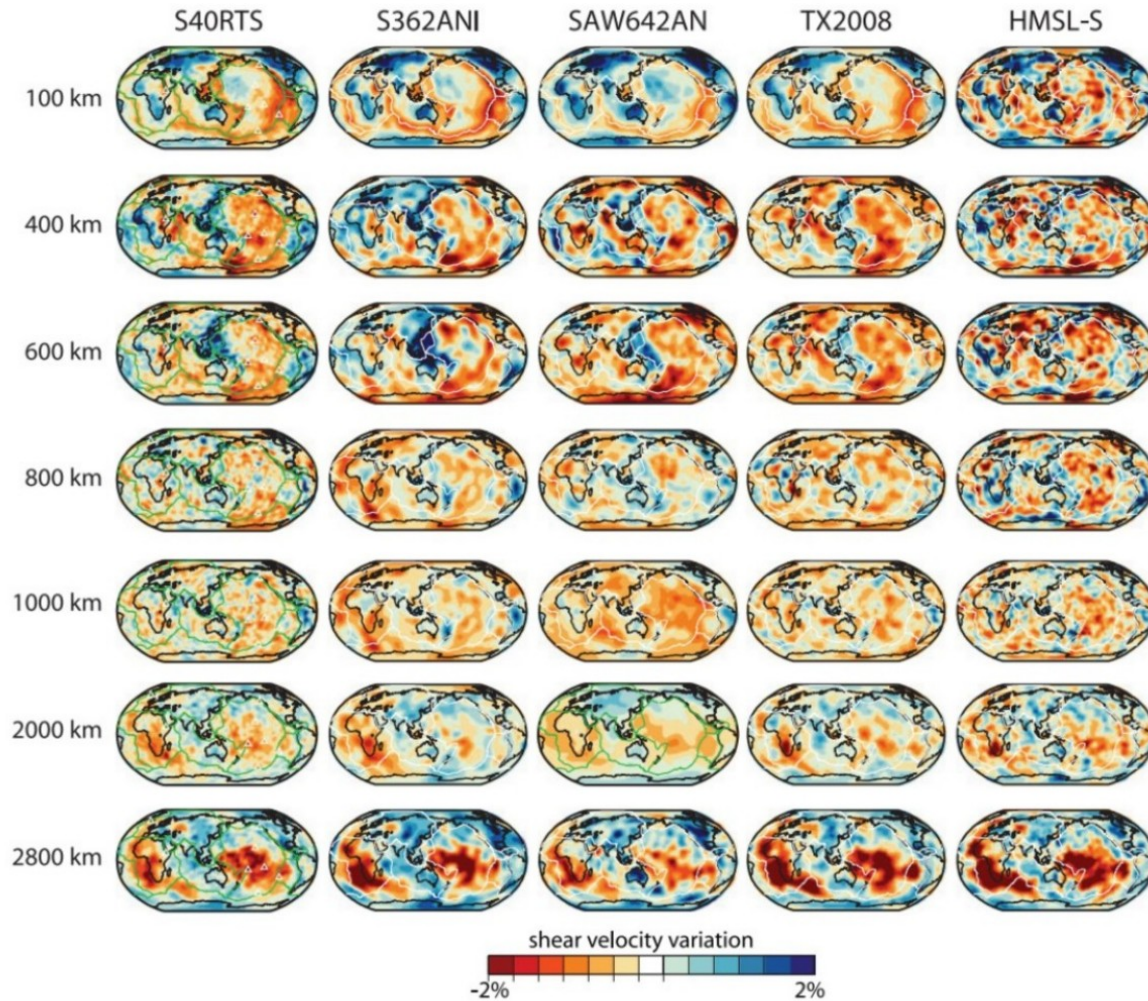
Woodhouse & Deuss (2007)

Using first-order perturbation theory, sensitivity kernels for eigenfrequencies with respect to radial velocity and density perturbations can be determined



Eigenfrequencies along with mass and moment of inertia used to constrain the spherically symmetric density variation within the Earth

Lateral density variations



What are the large low velocity regions at the base of the mantle?

Thermal or compositional?

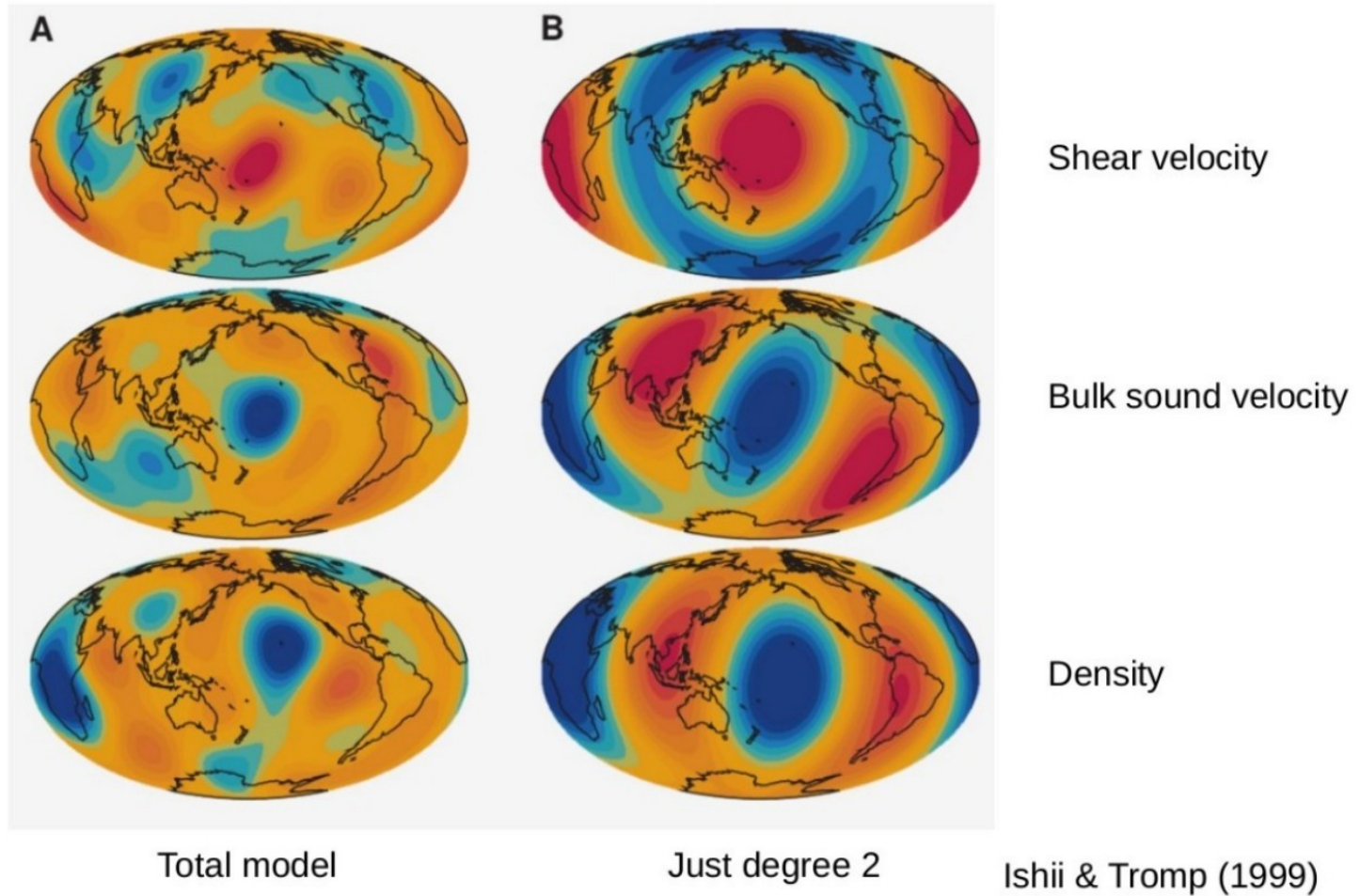
Bulk sound speed defined by:

$$\gamma = \sqrt{\frac{\kappa}{\rho}}.$$

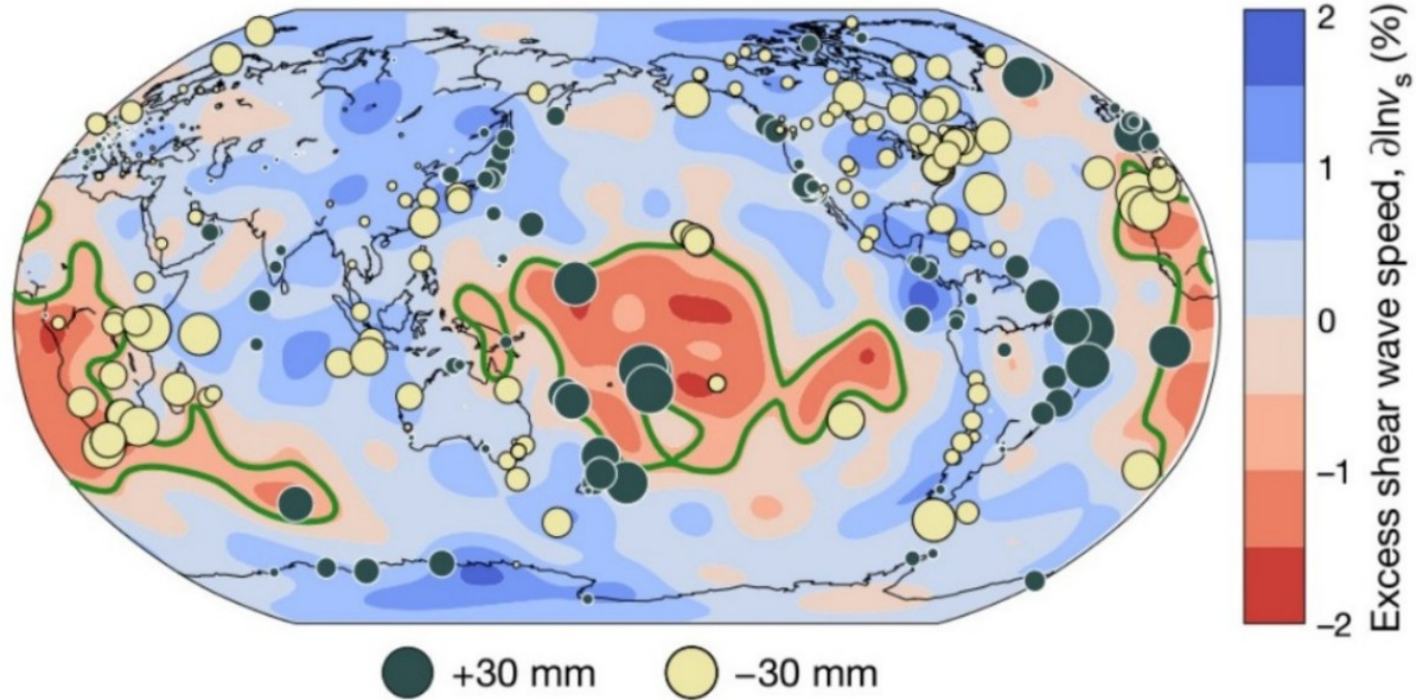
If velocity variations are due to **temperature alone**:

$$\frac{\delta\gamma}{\delta\beta} = \frac{\frac{\partial\gamma}{\partial T}}{\frac{\partial\beta}{\partial T}} > 0.$$

Joint P- and S-wave tomography suggest these velocities anti-correlate in the lower mantle – suggests a role for **compositional heterogeneity**



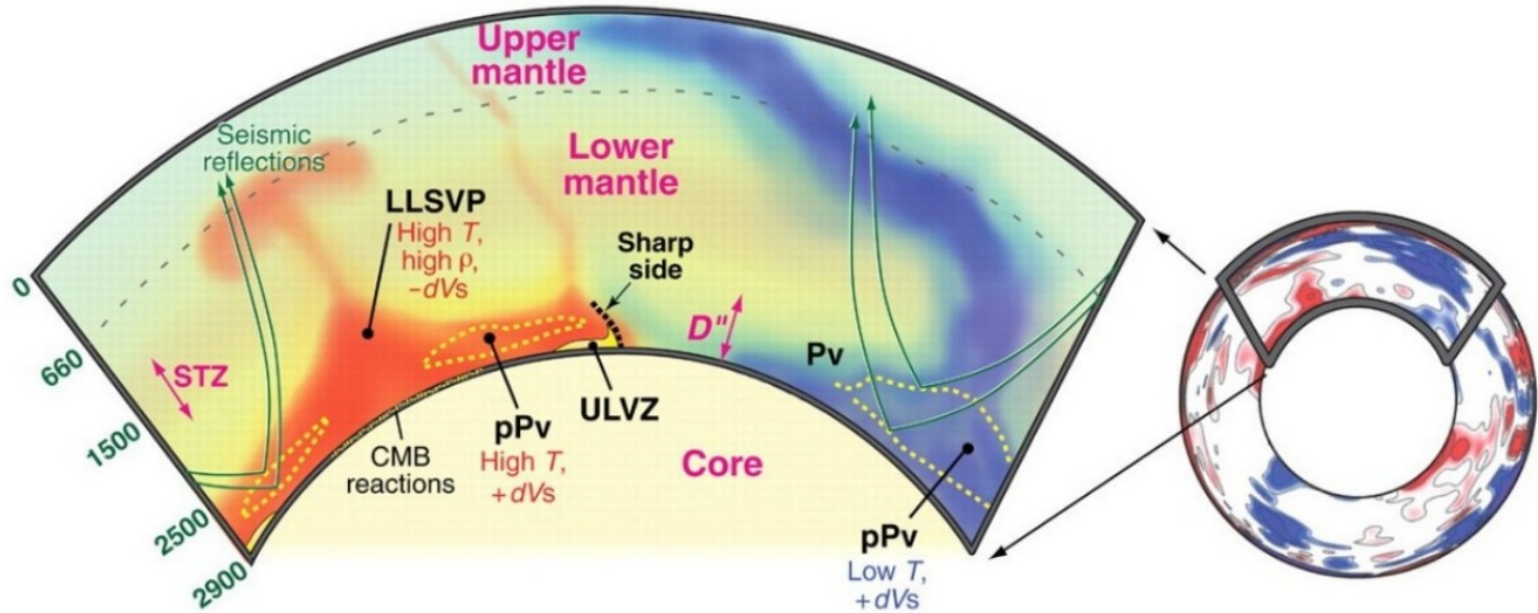
Early free oscillation tomography suggested LLSVPs are **dense features** – but conclusions have been questioned due to **approximations in forward modelling**



Lau *et al.* (2018)

Solid Earth tides also sensitive to deep mantle density. Results of a preliminary study are consistent with dense LLSVP

LLSVP origin?



Garnero & McNamara (2008)